AKTU

B.Tech 1st Year (Sem-1 / Sem-2)

Environment & Ecology

(BAS104 /BAS 204)

ONE SHOT

Unit-4

Current Environmental Issues

Environment and Ecology

AKTU Syllabus : Unit-4

Current Environmental Issues;

→ Global Warming

→ Green House Effects

→ Climate Change

→ Acid Rain

→ Ozone Layer Formation and Depletion

→ Population Growth and Automobile pollution

→ Burning of paddy straw

Environment and Ecology

UNIT-4 AKTU PYQs

$AKTU\ 2023-24\ Odd\ Sem$(1st Sem)

Q.1	*What are ozone – depleting substances? Give examples.*	***2 Marks***
Q.2	*What are the pieces of evidence of Climate Change?* *Also write the effects of acid rain.*	*7 Marks*
Q.3	*How does global warming affect climate change?* *Also write the effects of global warming.*	*7 Marks*
Q.4	*Write the solution of paddy straw burning. Also write the adverse effects on the environment of paddy straw burning.*	*7 Marks*

Environment and Ecology

UNIT-3 AKTU PYQs

$AKTU\ 2022-23\ Even\ Sem(2^{nd}\ sem)$

Q.1	*What is Global Warming? Name the gases responsible for global warming?*	**2 Marks**
Q.2	*What is Acid Rain? Name the gases responsible for acid rain? Discuss the causes, effects and remedies of acid rain on environment and ecology.*	7 Marks
Q.3	*What do you mean by the term Ozone hole? Discuss the reactions involved in the formation and depletion of ozone layer. Also explain the causes, effects and control of ozone layer depletion.*	7 Marks
Q.4	*What is Green House Effects? Name four green house gases. Discuss the causes, effects and remedies of green house effects.*	7 Marks

Environment and Ecology

GW GATEWAY CLASSES

UNIT-3 AKTU PYQs

AKTU 2022 – 23 Odd Sem (1st Sem)

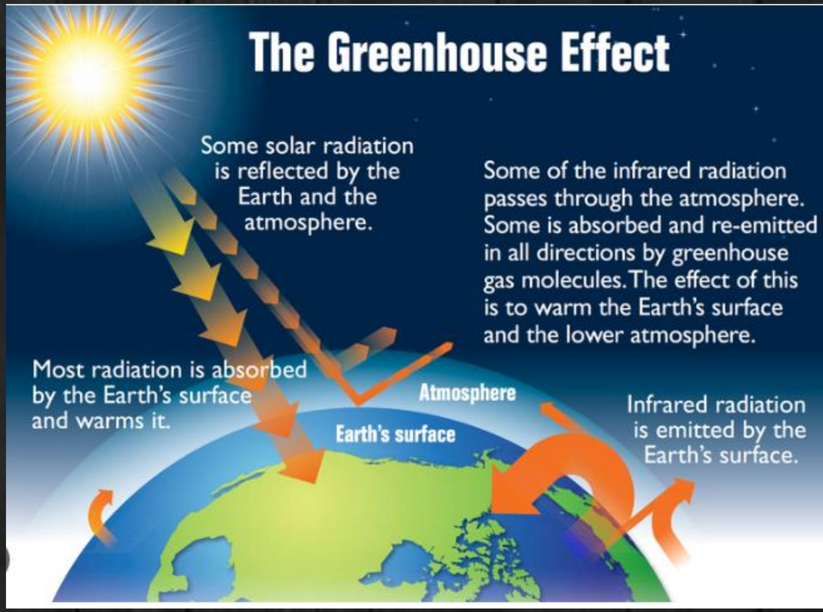
Q.1	*Explain the term 'population explosion'. Discuss its effects.*	**2 Marks**
Q.2	*What can be the effects of global warming? What are remedial measures? What are "Greenhouse Gases"? Discuss their contributions to global warming*	7 Marks
Q.3	*What are three major sources of emission in an automobile? Describe the direct control technologies that are used to reduce the emission rates from each of these sources.*	7 Marks
Q.4	*Differentiate between clean rain and acid rain. Name the principal species of acid rain. How they are formed and what are their effects?*	7 Marks

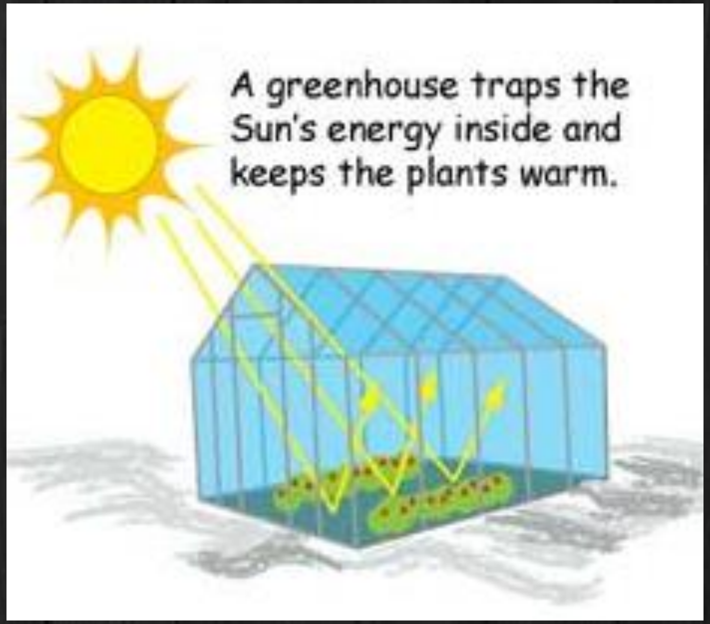
Green House Effects(GHE)

→ The greenhouse effect is a naturally occurring phenomenon that blankets the earth lower atmosphere and warms it, maintaining the temperature suitable for living things to survive.

→ Average global temperatures are maintained at about 15°C due to natural greenhouse effect.

→ Without this phenomenon, average global temperatures might have been around -17°C and at such low temperature life would not be able to exist.





Greenhouse Gasses (Gasses responsible for Greenhouse Effect/Global Warming):

→ Carbon Dioxide (CO_2)

→ Methane (CH_4)

→ Nitrous Oxide (N_2O)

→ Water vapor (H_2O)

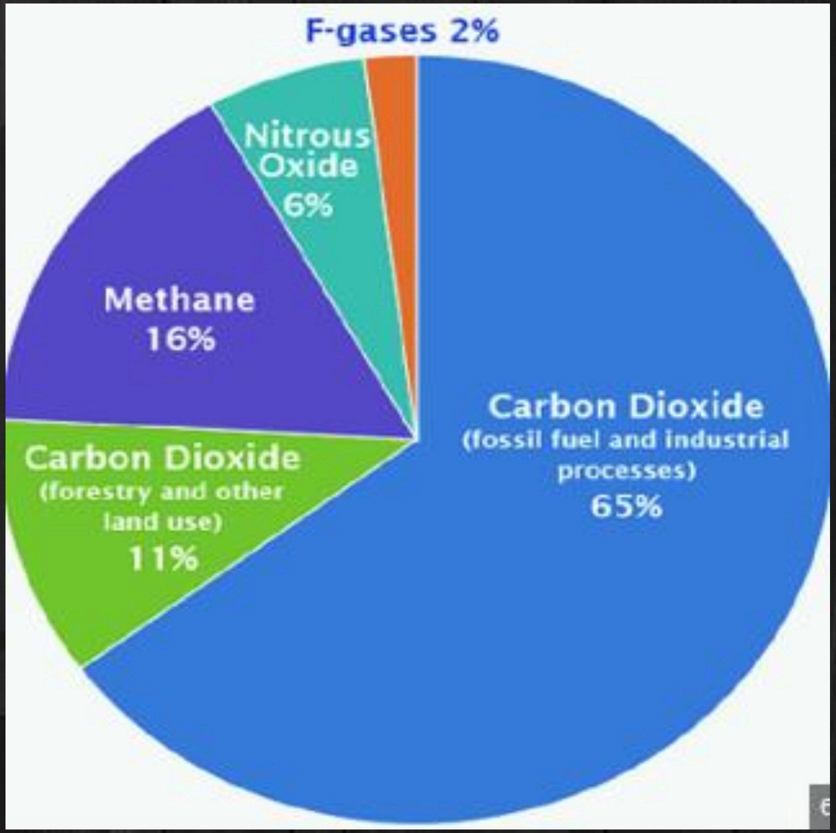
Fluorinated Gases

✓ Hydrofluorocarbons (HFCs)

✓ Perfluorocarbons (PFCs)

✓ Sulfur Hexafluoride (SF_6)

✓ Nitrogen Trifluoride (NF_3)



→ This effect is essential for life on Earth as it maintains the planet's temperature at a level suitable for living organisms.

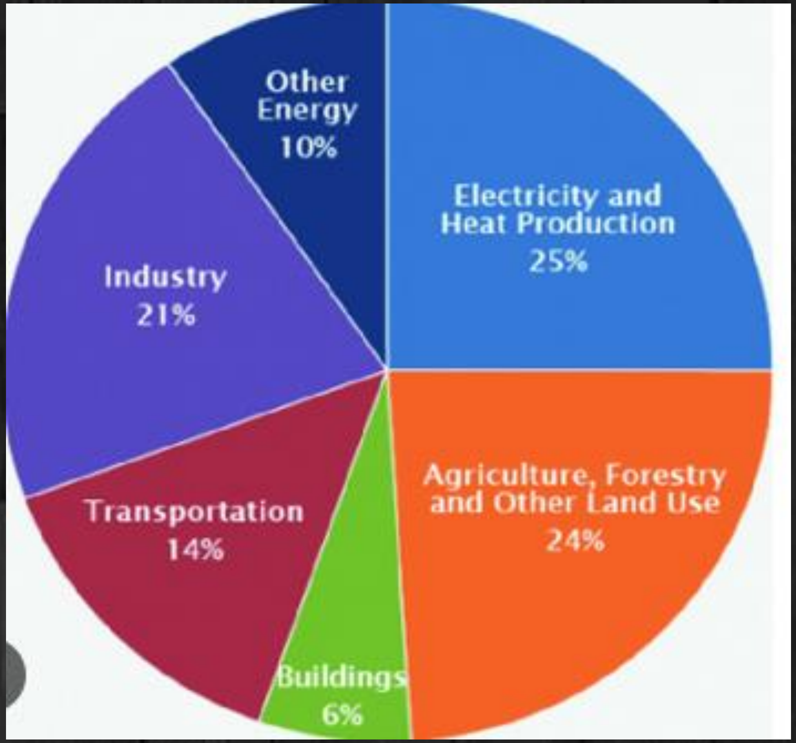
→ However, human activities have amplified this effect, leading to global warming and climate change.

Global Warming [It is Enhancement in GHE)

AKTU 2023 – 24 , 2022 – 23

→ Today when people talk about 'climate change', they mean the changes in climate over the last 100 years which is caused predominantly by human activity, especially Global Warming.

“Global warming is an average increase in the temperature of the atmosphere near the Earth's surface and in the troposphere, which can contribute to changes in global climate patterns”.



Causes of Global Warming/Enhancement in GHE

Electricity and Heat Production (25%):

→ Burning fossil fuels like coal, oil, and natural gas for electricity and heat generates significant CO_2 emissions.

Agriculture, Forestry, and Other Land Use (24%):

→ Agricultural activities release methane (CH_4) from livestock and nitrous oxide (N_2O) from fertilizers.

→ Deforestation reduces CO_2 absorption by trees, increasing atmospheric CO_2 levels.

Industry (21%):

→ Industrial processes and the use of fossil fuels for energy in manufacturing release CO_2 and other greenhouse gases.

Transportation (14%):

→ Emissions from cars, trucks, ships, trains, and planes contribute to CO_2 and other greenhouse gases due to the combustion of gasoline and diesel.

Other Energy (10%):

→ Includes emissions from energy extraction, refining, and distribution activities.

Buildings (6%):

→ Energy consumption in residential and commercial buildings for heating, cooling, lighting, and appliances leads to CO_2 emissions.

Effects of the Global Warming/Enhancement of GHE:

Global Warming: Increased average global temperatures.

Climate Change: Altered weather patterns, more intense storms, droughts, and heatwaves.

Melting Polar Ice: Rising sea levels due to melting glaciers and polar ice caps.

Ocean Acidification: Increased CO_2 levels lead to more acidic oceans, affecting marine life.

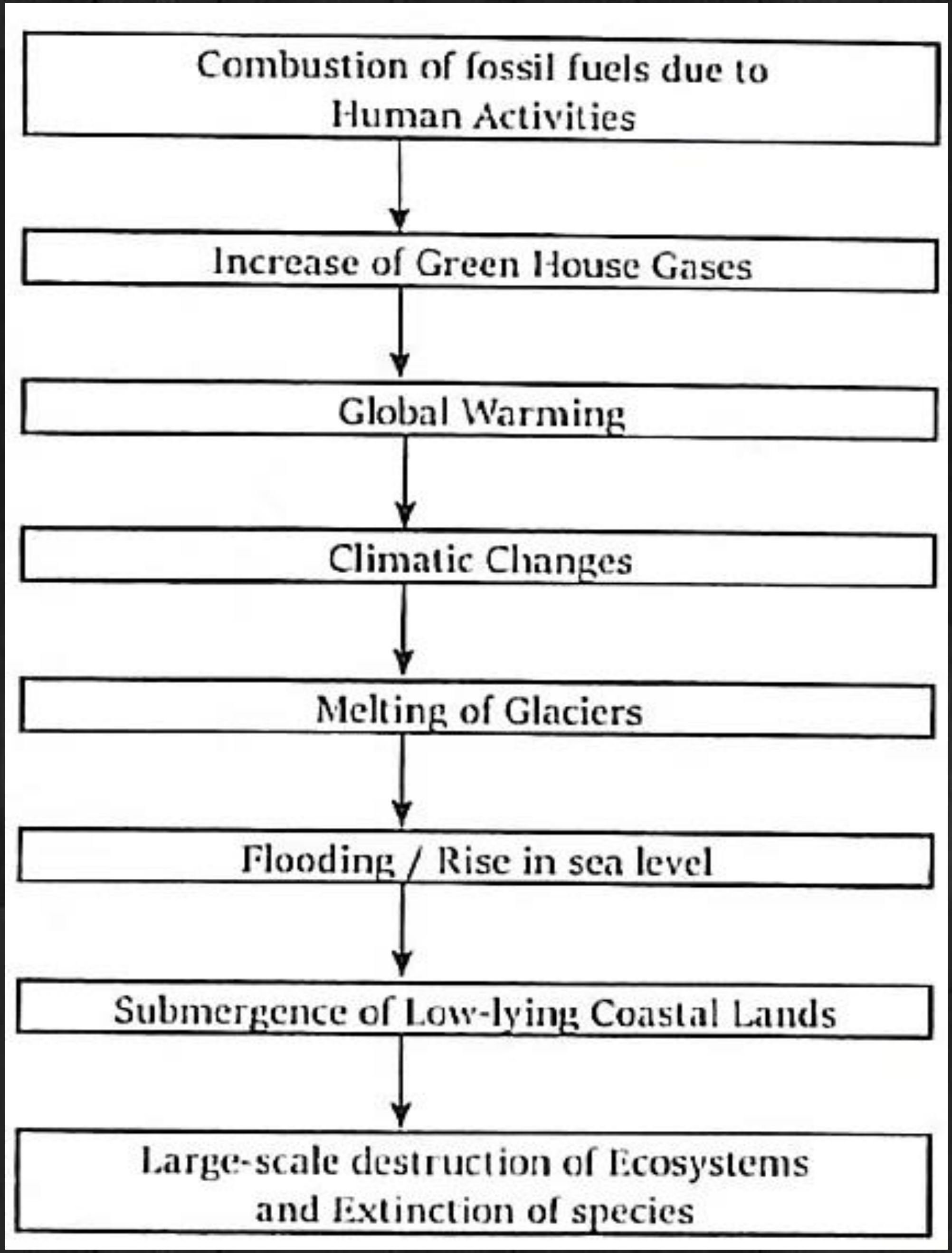
Remedies for the Global Warming/Enhancement of GHE:

Renewable Energy: Shifting to solar, wind, and other renewable energy sources to reduce fossil fuel usage.

Reforestation: Planting trees to absorb CO_2.

Energy Efficiency: Enhancing energy efficiency in buildings, transportation, and industries.

Reducing Emissions: Implementing policies and technologies to reduce greenhouse gas emissions.


Combustion of fossil fuels due to Human Activities
Increase of Green House Gases
Global Warming
Climatic Changes
Melting of Glaciers
Flooding / Rise in sea level
Submergence of Low-lying Coastal Lands
Large-scale destruction of Ecosystems and Extinction of species

Climate Change?

AKTU 2023 – 24

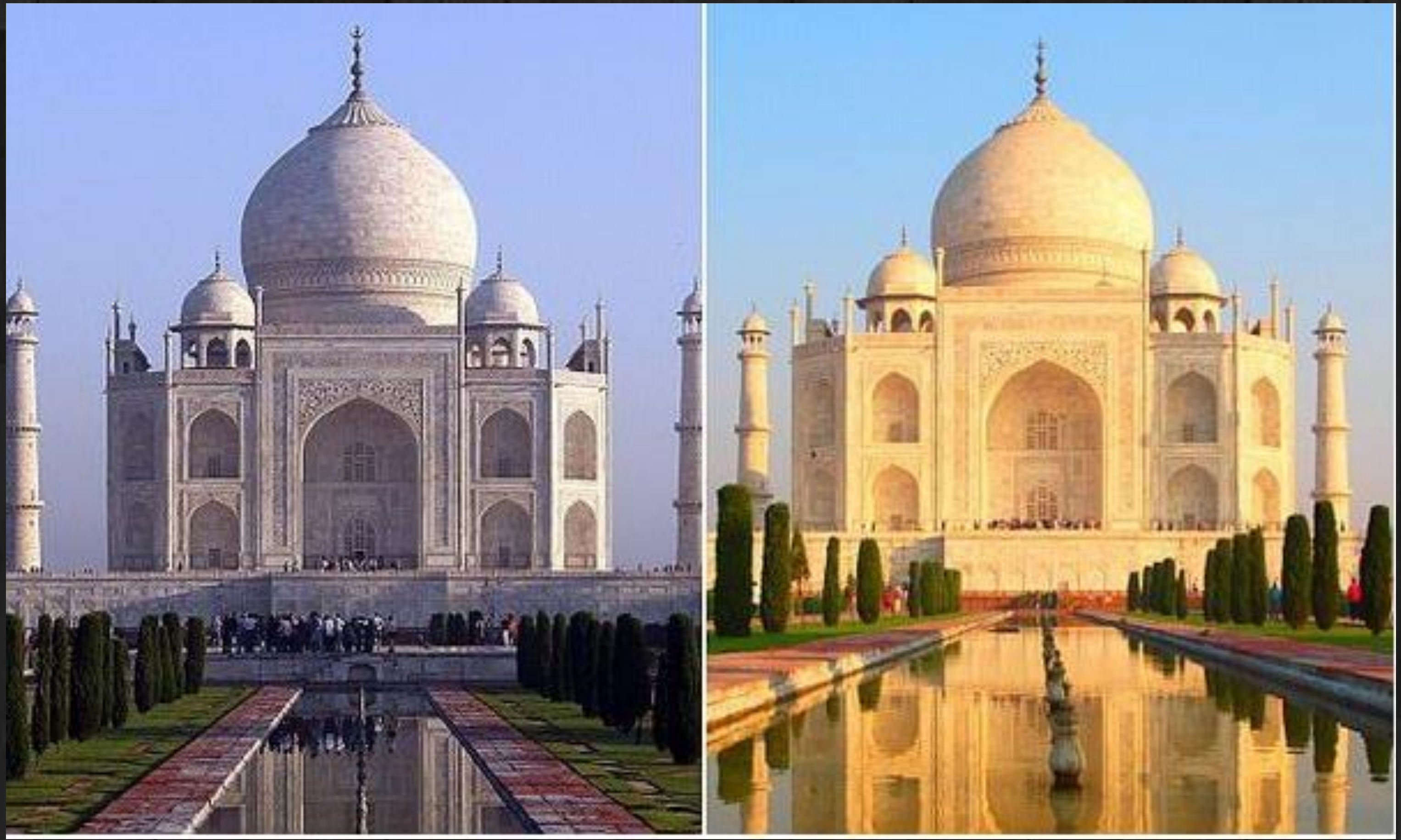
Climate change is ***significant changes*** in the Earth's climate patterns over an extended period. It includes changes in temperature, precipitation patterns, and more extreme weather events. Climate change can result from both natural processes and human activities.

Main Causes - Same as Global warming (Industrial processes, Deforestation, Agricultural practices,Burning of fossil fuels, Excessive transportation, Waste Management).

Effects of Climate change - Same as Global warming (leads to rising temperatures, melting ice, extreme weather events, ecosystem disruption, and economic and social impacts).

Mitigation involves reducing emissions, while adaptation focuses on managing the effects. Addressing climate change requires global cooperation and individual actions for a sustainable future.

Acid Rain



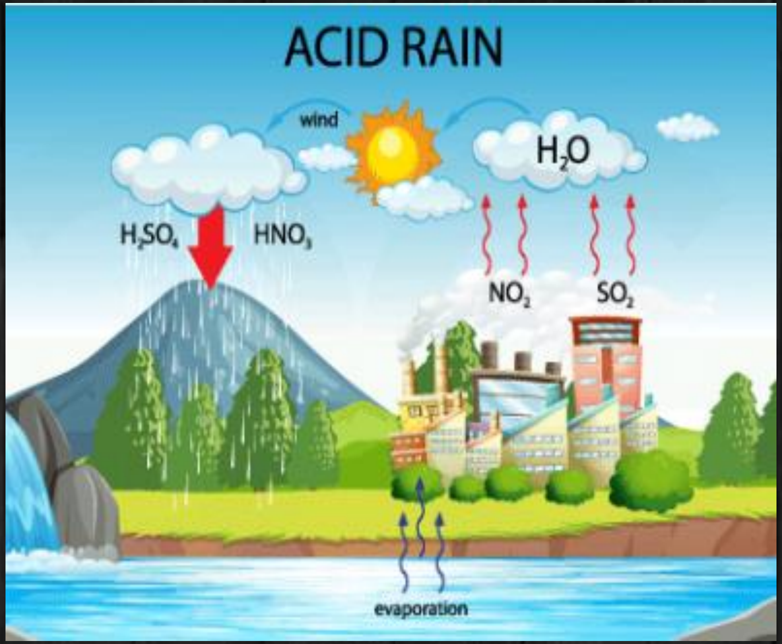
Acid Rain

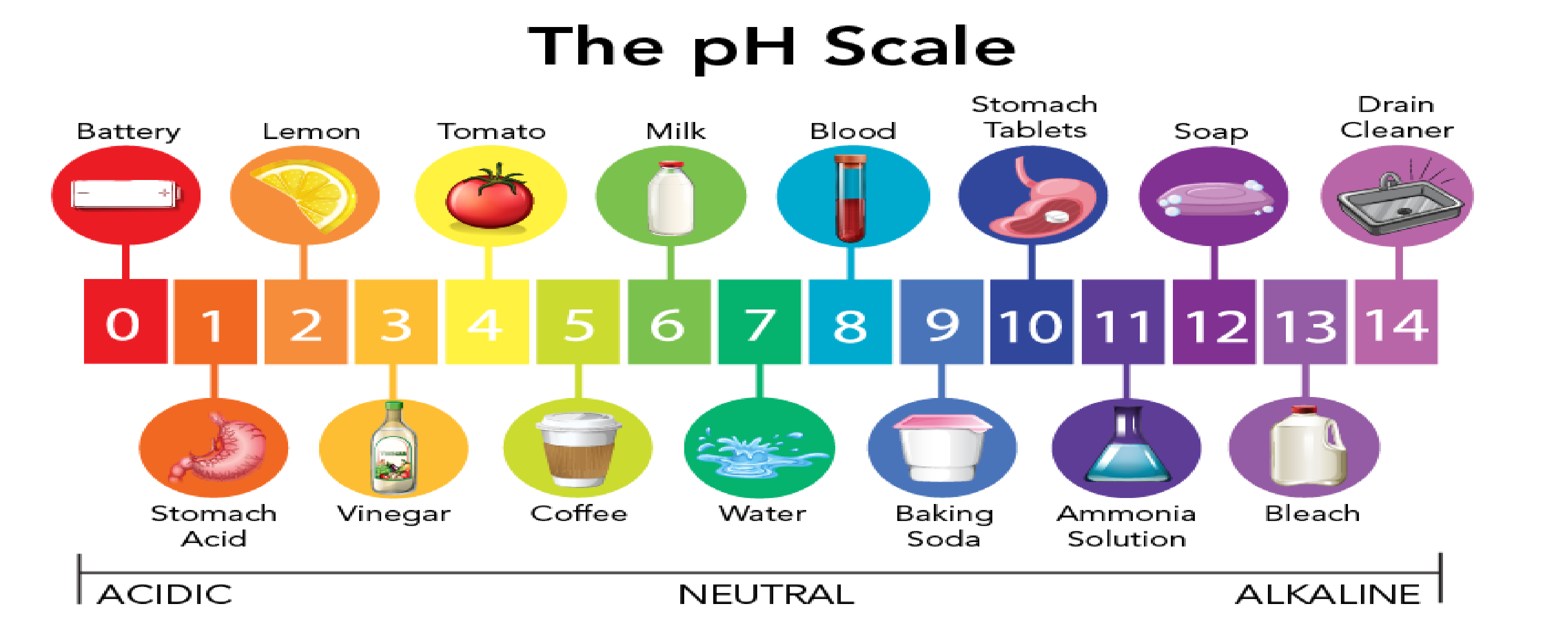
AKTU 2023 – 24 , 2022 – 23

→ Acid rain refers to any precipitation (rain, fog, mist, snow) that is more acidic than normal (pH of less than 5.6)

→ Acid rain is a broad term referring to a mixture of wet and dry deposition (a form of deposition material) from the atmosphere.

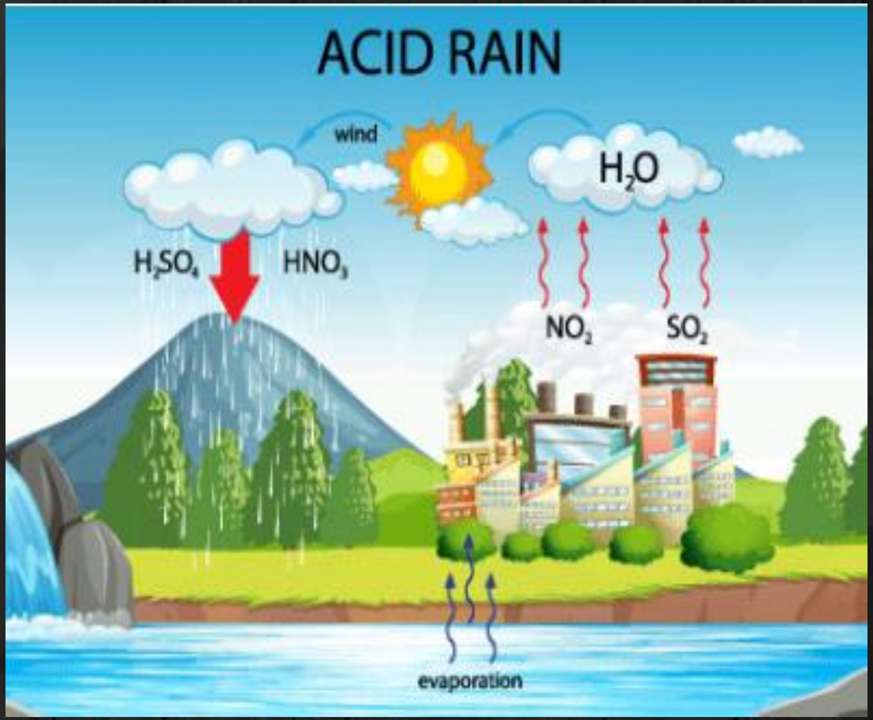
→ Primarily caused by the emission of sulfur dioxide (SO_2) and nitrogen oxides (NO_x) into the atmosphere from industrial activities, vehicles, and power plants.





Gases Responsible for Acid Rain

The primary gases responsible for acid rain are:



1.Sulfur Dioxide (SO_2):

Sources: Emitted from burning fossil fuels (coal, oil, natural gas) in power plants, industrial processes, and vehicles.

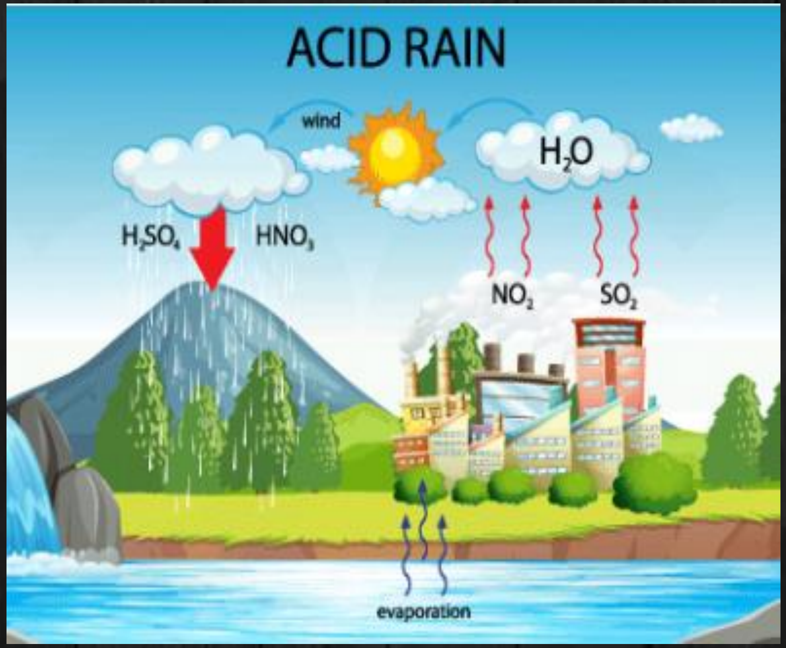
Reaction: Reacts with water vapor in the atmosphere to form sulfuric acid (H_2SO_4).

2. Nitrogen Oxides (NO_x):

Sources: Emitted from vehicle exhausts, industrial processes, and power plants.

Reaction: Reacts with water vapor in the atmosphere to form nitric acid (HNO_3).

Formation of Acid Rain



1.Emission of Pollutants:

→ Release SO_2 and NO_x into the atmosphere from industrial activities, vehicles, and power plants.

2.Chemical Reactions in the Atmosphere:

→ SO_2 reacts with water vapor and oxygen to form sulfuric acid (H_2SO_4).

→ NO_x reacts with water vapor and oxygen to form nitric acid (HNO_3).

3. Transport by Wind:

→ These acidic compounds can be transported over long distances by wind before falling to the ground.

4. Precipitation:

→ The sulfuric and nitric acids mix with cloud moisture and fall as acid rain, impacting forests, water bodies , and buildings.

Causes of Acid Rain

1. Burning of Fossil Fuels: Major source of SO_2 and NO_x emissions from power plants, vehicles, and industrial activities.

2. Industrial Processes: Emission of pollutants from factories and manufacturing units.

3. Agricultural Activities: Use of fertilizers and other practices that release ammonia, contributing to NO_x formation.

Effects of Acid Rain

1. Environmental Damage:

→ Acidifies water bodies, harming aquatic life.

→ Damages forests by leaching nutrients from the soil and affecting tree health.

2. Infrastructure Damage:

→ Corrodes buildings, monuments, and statues, particularly those made from limestone and marble.

→ Accelerates the decay of cultural heritage sites.

3. Human Health:

→ Contributes to respiratory illnesses, including asthma.

→ Indirectly affects human health through contamination of water and food sources.

Remedies of Acid rain

Reducing Emissions

→ Use of cleaner energy sources like wind, solar, and hydro power.

→ Implementation of cleaner technologies and practices in industrial processes.

Regulatory Actions:

→ Enforcing air quality standards and emission limits for sulfur dioxide and nitrogen oxides.

→ Promoting international cooperation to address transboundary air pollution.

Restoration Efforts:

→ Liming of acidic lakes and soils to neutralize acidity and restore ecosystems.

→ Reforestation and conservation efforts to protect and recover affected areas.

Differentiate Between Clean Rain and Acid Rain

Clean Rain:

Composition: Pure water with a neutral pH of about 5.6 due to the natural presence of carbon dioxide.

Sources: Natural evaporation and condensation cycles without significant pollution.

Acid Rain:

Composition: Water mixed with sulfuric and nitric acids, lowering the pH to below 5.6.

Sources: Polluted evaporation and condensation cycles, primarily due to emissions of sulfur dioxide and nitrogen oxides from human activities like burning fossil fuels and industrial processes.

Ozone Layer

AKTU 2022 – 23 even sem

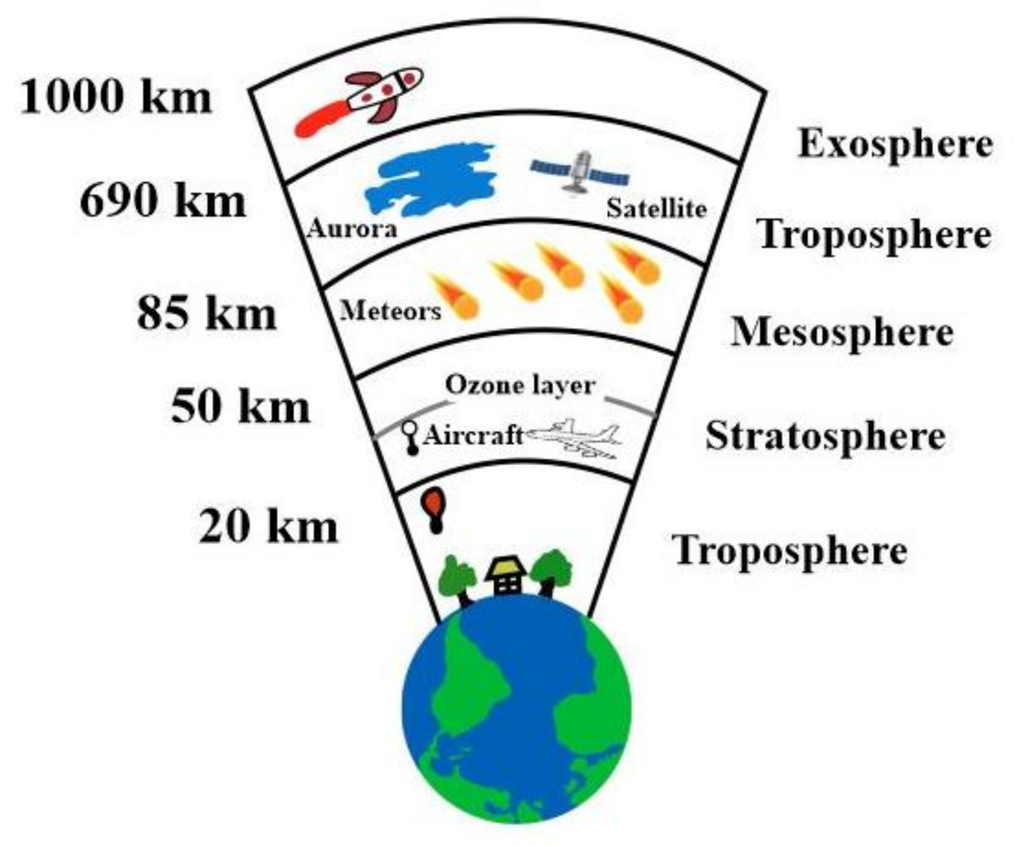
→ The ozone layer is a region of the Earth's stratosphere that contains a high concentration of ozone (O_3) molecules.

→ It is located about 10 to 30 kilometers above the Earth's surface and plays a crucial role in protecting life on Earth by absorbing the majority of the Sun's harmful ultraviolet (UV) radiation.

Ozone Hole?

→ The term "ozone hole" refers to the thinning of the ozone layer in the stratosphere, particularly over the Antarctic region, observed during the spring.

→ This phenomenon is primarily caused by human-made chemicals, such as chlorofluorocarbons (CFCs), which lead to the depletion of ozone molecules.



Ozone Layer Formation

Reactions Involved in the Formation:

Oxygen Molecule Photo-dissociation:

$$O_2 + UV \rightarrow 2\,O$$

→ High-energy UV radiation breaks down oxygen molecules (O_2) into individual oxygen atoms (O).

Ozone Formation:

$$O + O_2 \rightarrow O_3$$

→ The individual oxygen atoms (O) react with oxygen molecules (O_2) to form ozone (O_3).

Depletion of the Ozone Layer

→ Ozone layer depletion refers to the thinning and reduction of ozone concentration in the Earth's stratosphere.

→ This phenomenon has significant environmental and health impacts.

…Depletion of Ozone Layer

Chlorine and bromine atoms act as catalysts in the destruction of ozone. A single chlorine or bromine atom can destroy thousands of ozone molecules, leading to ozone depletion.

Reactions Involved in the Depletion of Ozone Layer

Release of CFCs and Other ODS (Ozone-Depleting Substances):

→ CFCs, halons, and other ODS released into the atmosphere slowly rise to the stratosphere.

Photo-dissociation of CFCs: $CFCl_3 + UV \rightarrow CFCl_2 + Cl$

→ Chlorofluorocarbons (CFCs) are broken down by ultraviolet (UV) light, releasing chlorine (Cl) atoms.

Ozone Depletion Reaction:

$Cl + O_3 \rightarrow ClO + O_2$

$ClO + O \rightarrow Cl + O_2$

The chlorine atoms react with ozone (O_3), forming chlorine monoxide (ClO) and molecular oxygen (O_2), and reducing the concentration of ozone.

Effects of Ozone Layer Depletion

Increased UV Radiation:

→ More ultraviolet (UV) radiation reaches the Earth's surface.

→ Leads to higher rates of skin cancer, and other health issues.

Impact on Ecosystems:

→ Marine life, especially phytoplankton, is adversely affected, disrupting the food chain.

→ Terrestrial plant life can suffer, affecting agriculture and natural vegetation.

Climate Change:

→ Ozone depletion contributes to changes in atmospheric circulation patterns, affecting climate.

Material Degradation:

→ Increased UV radiation can degrade materials like plastics, wood, fabrics, and rubber, reducing their lifespan.

Control Measures of ozone layer depletion

1.Montreal Protocol:

→ An international treaty agreed upon in 1987 to phase out the production and consumption of ozone-depleting substances (ODS).

→ It has been successful in reducing the global production of CFCs and other harmful chemicals.

2.Use of Alternatives:

→ Development and adoption of alternative substances that do not deplete the ozone layer, such as hydrofluorocarbons (HFCs) and other eco-friendly refrigerants.

3.Regulations and Legislation:

→ Implementation of strict regulations on the use of ODS in industries.

→ Encouraging industries to adopt best practices for handling and disposing of these chemicals.

4.Public Awareness and Education:

→ Increasing awareness about the importance of the ozone layer and ways to protect it.

→ Educating people on the proper disposal and recycling of products containing ODS.

Population Growth and Automobile Pollution

AKTU 2022 – 23 odd

Population Growth

→ Population growth refers to the increase in the number of individuals in a population.

Causes:

- **Birth Rate:** Higher birth rates contribute significantly to population growth.
- **Death Rate:** Advances in healthcare have reduced death rates, leading to an increase in population.
- **Immigration:** Movement of people from one region to another also affects population numbers.
- **Fertility Rates:** High fertility rates contribute to faster population growth.

Effects:

- **Resource Depletion:** Increased demand for natural resources such as water, minerals, and fossil fuels.
- **Environmental Degradation:** Overuse of land and deforestation to accommodate housing and agriculture.
- **Pollution:** Higher levels of waste and emissions from larger populations.
- **Strain on Infrastructure:** Overcrowding in cities, leading to insufficient infrastructure and services.

Population Growth Control:

Education: Promoting awareness about family planning and the impacts of overpopulation.

Healthcare: Improving access to contraception and reproductive health services.

Policies: Implementing policies to control birth rates, such as China's one-child policy.

Economic Development: Encouraging economic growth can lead to lower birth rates as seen in developed countries.

Automobile Pollution

→ Pollution caused by emissions from vehicles powered by internal combustion engines.

Causes:

Fossil Fuels: Burning of gasoline and diesel releases harmful pollutants.

Vehicle Numbers: Increasing number of vehicles on the road.

Traffic Congestion: Idling engines in traffic jams emit more pollutants.

Old Vehicles: Older vehicles often have less efficient engines and higher emissions.

Types of Pollutants:

Carbon Monoxide (CO): Poisonous gas that can cause health problems.

Nitrogen Oxides (NOx): Contribute to smog and acid rain.

Particulate Matter (PM): Tiny particles that can penetrate the lungs and cause respiratory issues.

Volatile Organic Compounds (VOCs): Contribute to the formation of ground-level ozone and smog.

Effects:

Health Problems: Respiratory issues, cardiovascular diseases, and other health problems.

Environmental Impact: Contributes to air pollution, acid rain, and climate change.

Economic Cost: Health care costs and loss of productivity due to pollution-related illnesses.

Automobile Pollution Control:

Regulations: Implementing stricter emission standards for vehicles.

Public Transport: Investing in efficient public transportation systems to reduce the number of private vehicles.

Electric Vehicles: Promoting the use of electric and hybrid vehicles that have lower emissions.

Technology: Encouraging the development of cleaner and more efficient engines.

Burning of Paddy Straw

→ Burning of paddy straw, also known as ***paddy residue*** or ***crop residue*** burning, refers to the ***practice of setting fire to the leftover plant material after harvesting rice crops.***

→ Burning the crop residue is a quick and inexpensive method to clear the fields for the next planting season.

Adverse Effects on the Environment

1. Air Pollution:

Emissions: Releases large amounts of pollutants, including carbon dioxide (CO_2), carbon monoxide (CO), methane (CH_4), volatile organic compounds (VOCs), and particulate matter (PM).

Health Impact: Causes respiratory problems, eye irritation, and other health issues for people living nearby.

2. Soil Degradation:

Nutrient Loss: Burning straw leads to the loss of essential soil nutrients like nitrogen, phosphorus, and potassium, reducing soil fertility.

Soil Microbes: High temperatures can kill beneficial soil microorganisms, affecting soil health and crop productivity.

3. Climate Change:

Greenhouse Gases: Contributes to global warming by releasing greenhouse gases like CO_2 and CH_4 into the atmosphere.

4. Visibility and Safety:

Smog: Smoke from burning straw can lead to the formation of smog, reducing visibility and causing safety hazards, particularly on roads.

Solutions to Paddy Straw Burning

1. Alternative Uses:

Composting: Converting paddy straw into compost to enrich the soil with organic matter and nutrients.

Bioenergy: Using paddy straw to produce bioenergy or biogas, providing a sustainable energy source and reducing pollution.

Animal Feed: Processing straw to use as fodder for livestock.

2. Technological Solutions:

Happy Seeder: A machine that allows direct sowing of wheat into the soil without removing the paddy straw, reducing the need for burning.

Baling Machines: Collect and compress straw into bales for easy transport and use in other applications.

3. Policy Measures:

Subsidies and Incentives: Providing financial incentives for farmers to adopt alternative methods and technologies for managing paddy straw.

Regulations: Enforcing stricter regulations and penalties against the open burning of paddy straw.

4. Awareness and Education:

Training Programs: Educating farmers on the adverse effects of burning straw and training them in sustainable farming practices.

Community Initiatives: Encouraging community-based initiatives and cooperation among farmers to find collective solutions for managing paddy straw.

Thank You